Wasserstein rate for finite irreducible aperiodic Markov chains from an arbitrary initial law

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Editorial note. The fixed source is arXiv 2401.15719v5; its footnote on page 1 states that this version corrects errors of the Math of OR published version, and the official v5 archive names its main file arxiv-v6.tex, which is an internal filename and not a different arXiv version.

Editorial note. Only the nonstationary aperiodic corollary 1125-1138 is counted here. The stationary theorem 946-962 is carried as context for the notation it fixes ('retain the notation from the stationary theorem') and no count is assigned to it in this card.

Editorial note. The abstract martingale theorem 199-281 and its complete proof 283-747, and the Hessian-weight variation lemma 818-942, are carried as supporting obligations because the proof applies them at 1226, 1333 and 1359; neither receives a separate count.

Editorial note. The letter h denotes both the Lipschitz test function and, in the Poisson-equation display at 785-788, the potential is named V; the extracted spans keep the source's own naming and no symbol has been renamed or repaired.

Allowed standard prerequisites (editorial).

- Finite-state Poisson equation and geometric ergodicity. For an irreducible aperiodic finite chain, V - PV = g is solvable for the centered reward g; the conditional covariance satisfies ||E(Sigma_k) - Sigma_infinity||_op <= C rho^(k-1) for some rho in (0,1).. Produces the potential V, the martingale differences m_k and the geometric tails behind the two-sided covariance sandwich. No rate is assumed.

- Vector martingale Wasserstein comparison. d_W(W_n, n^(-1/2)V_1^(1/2)Z) is bounded by two Taylor remainders multiplied by Tr(D_k) plus the signed covariance-interaction term Delta_n.. Applied to the martingale part after the boundary term is removed. Its signed interaction is exactly what forces the k-dependent centring in the nonstationary proof.

- Hessian-weight variation and Gaussian smoothing bounds. With H_k = V_k^(-1/2) A_{k,h}(F_(k-1)) V_k^(-1/2), the Mehler representation plus ||grad^j P_A h|| <= C||A^(-1/2)||^(j-1) for j = 2,3,4 gives sup_h sum_k E||H_k - H_(k-1)||_op <= C(1-beta)^(-2) n^((1-beta)/2).. Supplies the three spatial/internal/external comparison estimates that the

nonstationary proof re-derives from the covariance increments instead of stationarity.

- Matrix square root Lipschitz continuity and Gaussian coupling. On the positive-definite cone the square-root map is locally Lipschitz, so coupling n^(-1/2)V_1^(1/2)Z and Sigma_infinity^(1/2)Z through one standard normal vector costs O(n^(-1)).. Converts 1/n V_1 = Sigma_infinity + O(n^(-1)) into a Wasserstein comparison of the limiting Gaussian.

0.1 Preliminaries

In this subsection, we will describe the notation we use to establish the rate of convergence in the central limit theorem for vector-valued martingale differences. Let $\{m_k\}_{k\geq 1}$ be a d-dimensional martingale difference sequence with respect to a filtration $\{\mathcal{F}_k\}_{k\geq 0}$ i.e., m_k is $\mathcal{F}_k$-adapted, $E(||m_k||) < \infty$, where $||\cdot||$ is the standard Euclidean norm in $\Re^d$ and

$$E(m_k|\mathcal{F}_{k-1}) = 0 \qquad \forall k \geq 1.$$

Assumption 1. *Fix $\beta \in (0,1)$. We assume that the martingale difference sequence satisfies the following properties:*

1. *$E\|m_k\|^{2+\beta} < \infty$ for every $k \geq 1$.*
2. *$\Sigma_k = E(m_k m_k^T \mid \mathcal{F}_{k-1})$ exists for every $k \geq 1$.*

$\diamond$

Item (1) in the assumption is motivated by the fact that we want to extend the techniques in (Gallouët et al. 2018) to martingales .

Define $S_n = \sum_{k=1}^n m_k$. We are interested in whether the following normalized sum

$$W_n := \frac{S_n}{\sqrt{n}} \tag{1}$$

converges to $N(0, \Sigma_\infty)$, and if so, we are interested in the rate of convergence. We will use Stein's method to establish such a convergence and to estimate the rate of convergence (Stein 1972, Ross 2011, Chatterjee 2014).

The Wasserstein distance between two d-dimensional random vectors X and Y is defined as

$$d_{\mathcal{W}}(X,Y) = \sup_{h \in Lip_1} E[h(X) - h(Y)],$$

where Lip_L, the class of Lipschitz functions from $\Re^d$ to $\Re$ with Lipschitz constant L, is defined as

$$Lip_L = \{h : |h(x) - h(y)| \leq L||x-y||\}.$$

For later reference, we use the notation $||A||_{op}$ to denote the operator norm of a matrix A, i.e.,

$$||A||_{op} = \sup_{x:||x||=1} ||Ax||.$$

The following results from (Gallouët et al. 2018, Propositions 2.2 and 2.3) and (Fang et al. 2019, Theorem 3.1) will be useful to us and so we present them together as a lemma for future reference.

Lemma 1. *Let $g \in Lip_L$ and f be the solution to Stein's equation:*

$$g(u) - E(g(Z)) = \Delta f(u) - u^T \nabla f(u), \tag{2}$$

where Δ and ∇ are the Laplacian and gradient operators, respectively, and $Z \sim N(0, I)$. Then, f has the following regularity properties for any $\beta \in (0, 1)$:

1. $||\nabla^2 f(x) - \nabla^2 f(y)||_{op} \leq \tilde{C}_1(d, \beta)||x - y||^\beta L$, *where ∇^2 is the Hessian operator.*
2. $|\Delta f(x) - \Delta f(y)| \leq \tilde{C}_2(d, \beta)||x - y||^\beta L$, *where*

$$\tilde{C}_1 = C_1(d) + \frac{2}{1-\beta}, \quad \tilde{C}_2 = C_2(d) + \frac{2d}{1-\beta}, \quad C_1 = 2^{3/2}\frac{(1+2d)\Gamma((1+d)/2)}{d\Gamma(d/2)}, \quad C_2 = 2\sqrt{\frac{2d}{\pi}}.$$

3. $$\left|\langle \nabla^2 f(x), u_1 u_2^T \rangle_{HS}\right| \leq CL||u_1|| \cdot ||u_2|| \, \forall x, u_1, u_2 \in \Re^d,$$

   where C is a constant (independent of d and β) and $\langle \cdot, \cdot \rangle_{HS}$ is the Hilbert-Schmidt inner product (also called the Frobenius inner product), i.e., the sum of the element-by-element product of two matrices.

$\diamond$

1 Markov Chain Central Limit Theorem

Consider a time-homogeneous Markov chain $\{X_k\}_{k \geq 0}$ on a state space $\mathcal{S}$ with initial distribution η. Let $r(X_k) \in \Re^d$ denote a vector reward and let $\bar{r} = \pi r$ whenever the chain has invariant distribution π. We are interested in rates of convergence to normality of the centered additive functional

$$\frac{1}{\sqrt{n}} \sum_{k=1}^{n} \big(r(X_{k-1}) - \bar{r}\big).$$

As is common in the literature, we study this scaled sum by relating the reward process to a martingale via Poisson's equation (Douc et al. 2018, Makowski and Shwartz 2002, Glynn and Meyn 1996, Glynn and Infanger 2023).

1.1 Finite State Space Markov Chains

We now show that the martingale theorem can be used to obtain a non-asymptotic CLT for finite-state Markov chains. In Theorem 2, one must control the signed covariance-interaction term. This can be done by applying a second Poisson equation, now to the conditional covariance function.

Throughout this subsection, assume that the Markov chain is stationary, finite-state, and irreducible. Periodicity is allowed. Let

$$g(x) := r(x) - \bar{r}, \qquad \bar{r} := \pi r.$$

Let V solve the Poisson equation

$$V - PV = g.$$

Then

$$g(X_{k-1}) = V(X_{k-1}) - V(X_k) + m_k,$$

where

$$m_k := V(X_k) - E[V(X_k) \mid X_{k-1}]$$

is a martingale difference sequence. Define

$$\Sigma_k := E[m_k m_k^\top \mid \mathcal{F}_{k-1}] = v(X_{k-1}),$$

where

$$v(x) := E[m_1 m_1^\top \mid X_0 = x].$$

Since the chain is stationary,

$$E(\Sigma_k) = \pi(v) =: \Sigma_\infty.$$

Assume throughout this subsection that Σ_∞ is positive definite.

Theorem 1. *Let*

$$U_n = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} (r(X_{k-1}) - \bar{r}).$$

Assume that the Markov chain is stationary, finite-state, and irreducible, and that $\Sigma_\infty \succ 0$. Then, for every $\beta \in (0,1)$,

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le \frac{C}{(1-\beta)^2} n^{-\beta/2},$$

where C does not depend on n or β. Consequently,

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) = O\left(\frac{(\log n)^2}{\sqrt{n}}\right).$$

Corollary 1. *Assume that the Markov chain is finite-state, irreducible, and aperiodic, with an arbitrary initial distribution. If $\Sigma_\infty \succ 0$, then, for every $\beta \in (0,1)$,*

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le \frac{C}{(1-\beta)^2} n^{-\beta/2},$$

with C independent of n and β, and consequently

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) = O\left(\frac{(\log n)^2}{\sqrt{n}}\right).$$

Theorem 2. *Let*

$$V_k := \sum_{j=k}^{n} E(\Sigma_j), \qquad k = 1, \dots, n,$$

and assume that V_k is positive definite for each $k = 1, \dots, n$. Define

$$D_k := V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}, \qquad k = 1, \dots, n.$$

Let $Z \sim N(0, I)$, and let W_n be defined as in (1). Then, under Assumption 1 for the same fixed $\beta \in (0, 1)$,

$$d_{\mathcal{W}}(W_n, n^{-1/2}V_1^{1/2}Z) \le \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \Bigg[\tilde{C}_1(d,\beta) \|V_k^{1/2}\|_{op} E\left(\|V_k^{-1/2} m_k\|^{2+\beta} \right) + \tilde{C}_1(d,\beta) \operatorname{Tr}(D_k) \|V_k^{1/2}\|_{op} E\left(\|V_k^{-1/2} m_k\|^{\beta} \right) \Bigg] + \Delta_n,$$

where

$$\Delta_n := \sup_{h \in Lip_1} \frac{1}{\sqrt{n}} \left| \sum_{k=1}^{n} E\left[\operatorname{Tr}\left(A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2} \left(E(\Sigma_k) - \Sigma_k \right) V_k^{-1/2} \right) \right] \right|.$$

Here

$$A_{k,h}(\mathcal{F}_{k-1}) := E\left[\nabla^2 f_{k,h}(\widetilde{Z}_k) \mid \mathcal{F}_{k-1} \right],$$

where $f_{k,h}$ is the $\mathcal{F}_{k-1}$-measurable random solution of the Stein equation defined in the proof below with

$$a = S_{k-1}, \qquad Q_k = V_k^{1/2}.$$

and

$$\widetilde{Z}_k := V_k^{-1/2} T_{k+1} \sim N\left(0, V_k^{-1/2} V_{k+1} V_k^{-1/2} \right),$$

with the convention $V_{n+1} = 0$. Moreover,

$$\|A_{k,h}(\mathcal{F}_{k-1})\|_{op} \le C \|V_k^{1/2}\|_{op}, \quad a.s.,$$

where C is the constant in statement (3) of Lemma 1. Since $0 \preceq D_k \preceq I$, the same bound also implies the coarser version obtained by replacing $\operatorname{Tr}(D_k)$ by d. The sharper form above is used in the Markov-chain applications.

Proof. Fix $h \in Lip_1$. By the standard rescaling of Lipschitz test functions,

$$d_{\mathcal{W}}(W_n, n^{-1/2}V_1^{1/2}Z) = \sup_{h \in Lip_1} \frac{1}{\sqrt{n}} E\left[h(S_n) - h(T_1) \right],$$

where

$$T_k := \sum_{j=k}^{n} Y_j,$$

and $Y_1, \ldots, Y_n$ are independent Gaussian random vectors, independent of the martingale difference sequence, with

$$Y_j \sim N(0, E(\Sigma_j)).$$

Thus

$$T_k \sim N(0, V_k).$$

The rescaling follows because if $g \in Lip_1$, then $h(x) := \sqrt{n}\, g(x/\sqrt{n})$ also belongs to Lip_1, and

$$E\left[g(W_n) - g(n^{-1/2}V_1^{1/2}Z) \right] = \frac{1}{\sqrt{n}} E\left[h(S_n) - h(T_1) \right].$$

Using the Lindeberg decomposition,

$$E\left[h(S_n) - h(T_1)\right] = \sum_{k=1}^{n} E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k)\right],$$

where $T_{n+1} = S_0 = 0$.

We now bound a single summand. Fix k. Conditional on $\mathcal{F}_{k-1}$, set

$$a = S_{k-1}, \qquad Q_k = V_k^{1/2}.$$

Define

$$\widetilde{h}_k(u) := h(a + Q_k u).$$

Since $h \in Lip_1$, we have

$$\widetilde{h}_k \in Lip_{L_k}, \qquad L_k = \|Q_k\|_{op} = \|V_k^{1/2}\|_{op}.$$

Let $f_k = f_{k,h}$ be the solution of the Stein equation

$$\widetilde{h}_k(u) - E[\widetilde{h}_k(Z)] = \Delta f_k(u) - u^\top \nabla f_k(u).$$

Since $a = S_{k-1}$ is $\mathcal{F}_{k-1}$-measurable, the function f_k is also $\mathcal{F}_{k-1}$-measurable as a random function.

By the Stein equation, with

$$\widetilde{Z}_k := V_k^{-1/2} T_{k+1},$$

we have

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&= E\left[\Delta f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) - \left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)^\top \nabla f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Moreover,

$$\widetilde{Z}_k \sim N(0, \Lambda_k), \qquad \Lambda_k := V_k^{-1/2} V_{k+1} V_k^{-1/2}.$$

Since

$$V_k = E(\Sigma_k) + V_{k+1},$$

we have

$$I - \Lambda_k = V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}.$$

We rewrite the preceding conditional expectation as

$$\begin{aligned}
&E\left[\mathrm{Tr}\left(\Lambda_k \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) - \widetilde{Z}_k^\top \nabla f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) \mid \mathcal{F}_{k-1}\right] \\
&+ E\left[\mathrm{Tr}\left((I - \Lambda_k) \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) \mid \mathcal{F}_{k-1}\right] \\
&- E\left[(V_k^{-1/2} m_k)^\top \nabla f_k(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}\right] \\
&- E\left[(V_k^{-1/2} m_k)^\top \left(\nabla f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) - \nabla f_k(\widetilde{Z}_k)\right) \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

The first term is zero by Gaussian integration by parts: conditional on $\mathcal{F}_k$, the quantity $V_k^{-1/2} m_k$ is fixed and $\widetilde{Z}_k \sim N(0, \Lambda_k)$ is independent of $\mathcal{F}_k$. The third term is zero because f_k is $\mathcal{F}_{k-1}$-measurable as a random function, $\widetilde{Z}_k$ is independent of $\mathcal{F}_k$, and $E[m_k \mid \mathcal{F}_{k-1}] = 0$.

Using the fundamental theorem of calculus, we obtain

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&= E\left[\mathrm{Tr}\left(V_k^{-1/2} E(\Sigma_k) V_k^{-1/2} \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) \mid \mathcal{F}_{k-1}\right] \\
&\quad - E\left[(V_k^{-1/2} m_k)^\top \left(\nabla^2 f_k(\Theta V_k^{-1/2} m_k + \widetilde{Z}_k) - \nabla^2 f_k(\widetilde{Z}_k)\right)(V_k^{-1/2} m_k) \mid \mathcal{F}_{k-1}\right] \\
&\quad - E\left[(V_k^{-1/2} m_k)^\top \nabla^2 f_k(\widetilde{Z}_k)(V_k^{-1/2} m_k) \mid \mathcal{F}_{k-1}\right],
\end{aligned}$$

where $\Theta \sim \mathrm{Uniform}[0,1]$ is independent of all other random variables.

Define

$$A_{k,h}(\mathcal{F}_{k-1}) := E\left[\nabla^2 f_k(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}\right].$$

Using

$$\Sigma_k = E[m_k m_k^\top \mid \mathcal{F}_{k-1}],$$

the last term becomes

$$-\mathrm{Tr}\left(V_k^{-1/2} \Sigma_k V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1})\right).$$

By statement (1) of Lemma 1, the second term above is bounded in absolute value by

$$\tilde{C}_1(d,\beta) \|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right].$$

Therefore,

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&\le E\left[\mathrm{Tr}\left(V_k^{-1/2} E(\Sigma_k) V_k^{-1/2} \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) \mid \mathcal{F}_{k-1}\right] \\
&\quad - \mathrm{Tr}\left(V_k^{-1/2} \Sigma_k V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1})\right) \\
&\quad + \tilde{C}_1(d,\beta) \|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Next, add and subtract

$$\mathrm{Tr}\left(V_k^{-1/2} E(\Sigma_k) V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1})\right).$$

This gives

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&\le \mathrm{Tr}\left(A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2} \left(E(\Sigma_k) - \Sigma_k\right) V_k^{-1/2}\right) \\
&\quad + E\left[\mathrm{Tr}\left(D_k\left[\nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) - \nabla^2 f_k(\widetilde{Z}_k)\right]\right) \mid \mathcal{F}_{k-1}\right] \\
&\quad + \tilde{C}_1(d,\beta) \|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right],
\end{aligned}$$

where

$$D_k := V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}.$$

It remains to bound the weighted trace term. Since

$$V_k = E(\Sigma_k) + V_{k+1},$$

we have

$$D_k = I - V_k^{-1/2} V_{k+1} V_k^{-1/2}.$$

Hence

$$0 \preceq D_k \preceq I.$$

Therefore, for any $x, u \in \mathbb{R}^d$,

$$\begin{aligned}
&\left|\operatorname{Tr}\left(D_k\left[\nabla^2 f_k(x+u) - \nabla^2 f_k(x)\right]\right)\right| \\
&\leq \operatorname{Tr}(D_k)\left\|\nabla^2 f_k(x+u) - \nabla^2 f_k(x)\right\|_{op} \\
&\leq \operatorname{Tr}(D_k)\,\tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op}\|u\|^\beta.
\end{aligned}$$

Applying this bound with

$$x = \widetilde{Z}_k, \qquad u = V_k^{-1/2} m_k,$$

and taking conditional expectation gives

$$\begin{aligned}
&\left|E\left[\operatorname{Tr}\left(D_k\left[\nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) - \nabla^2 f_k(\widetilde{Z}_k)\right]\right) \mid \mathcal{F}_{k-1}\right]\right| \\
&\leq \tilde{C}_1(d,\beta)\operatorname{Tr}(D_k)\|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^\beta \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Thus,

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&\leq \operatorname{Tr}\left(A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2}\left(E(\Sigma_k) - \Sigma_k\right) V_k^{-1/2}\right) \\
&\quad + \tilde{C}_1(d,\beta)\operatorname{Tr}(D_k)\|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^\beta \mid \mathcal{F}_{k-1}\right] \\
&\quad + \tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Taking expectations, summing over k, substituting into the Lindeberg decomposition, and using the rescaling identity for the Wasserstein distance, we obtain, for the fixed $h \in Lip_1$,

$$\begin{aligned}
&E\left[h(W_n) - h(n^{-1/2} V_1^{1/2} Z)\right] \\
&\leq \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \Bigg[\tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op} E\left(\|V_k^{-1/2} m_k\|^{2+\beta}\right) \\
&\qquad\qquad + \tilde{C}_1(d,\beta)\operatorname{Tr}(D_k)\|V_k^{1/2}\|_{op} E\left(\|V_k^{-1/2} m_k\|^\beta\right)\Bigg] \\
&\quad + \frac{1}{\sqrt{n}}\left|\sum_{k=1}^{n} E\left[\operatorname{Tr}\left(A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2}\left(E(\Sigma_k) - \Sigma_k\right) V_k^{-1/2}\right)\right]\right|.
\end{aligned}$$

Taking the supremum over $h \in Lip_1$ proves the claimed bound.

Finally, statement (3) of Lemma 1 gives

$$\|A_{k,h}(\mathcal{F}_{k-1})\|_{op} \le C\|V_k^{1/2}\|_{op}.$$

This completes the proof. □

Lemma 2. *Let*

$$V_k := \sum_{j=k}^{n} E(\Sigma_j) = (n-k+1)\Sigma_\infty, \qquad k = 1, \dots, n,$$

and

$$H_k := V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2},$$

where

$$A_{k,h}(\mathcal{F}_{k-1}) := E[\nabla^2 f_{k,h}(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}]$$

is the matrix appearing in Theorem 2. Then, for every $\beta \in (0,1)$,

$$\sup_{h \in Lip_1} \sum_{k=2}^{n} E\|H_k - H_{k-1}\|_{op} \le \frac{C}{(1-\beta)^2} n^{(1-\beta)/2},$$

where C depends on the chain, r, d, and Σ_∞, but not on n or β.

Proof. Fix $h \in Lip_1$. Let $c_t := \sqrt{1-e^{-2t}}$, and let

$$P_A h(x) := E[h(x + A^{1/2}Z)], \qquad Z \sim N(0, I),$$

denote Gaussian convolution with covariance A. The Mehler representation of the standard normal Stein solution, followed by differentiation under the integral, gives

$$H_k = -\int_0^\infty e^{-2t} E_{\xi \sim N(0, V_{k+1})}[\nabla^2 P_{c_t^2 V_k} h(S_{k-1} + e^{-t}\xi)]\, dt. \tag{3}$$

The differentiation is justified by the standard Gaussian-smoothing bounds

$$\|\nabla^j P_A h\|_{op} \le C\|A^{-1/2}\|_{op}^{j-1}, \qquad j = 2, 3, 4.$$

Here $\|\cdot\|_{op}$ denotes the usual matrix operator norm for matrices and Hessians, and the multilinear operator norm for high-order derivatives.

The second- and third-derivative bounds imply, respectively,

$$\|\nabla^2 P_A h(x) - \nabla^2 P_A h(y)\|_{op} \le 2C\|A^{-1/2}\|_{op},$$
$$\|\nabla^2 P_A h(x) - \nabla^2 P_A h(y)\|_{op} \le C\|A^{-1/2}\|_{op}^2 \|x - y\|.$$

Since $\min\{1, u\} \le u^\beta$ for $u \ge 0$, interpolation gives

$$\|\nabla^2 P_A h(x) - \nabla^2 P_A h(y)\|_{op} \le C\|A^{-1/2}\|_{op}^{1+\beta} \|x - y\|^\beta, \tag{4}$$

with C uniform over $\beta \in (0,1)$.

Because the state space is finite, m_k is uniformly bounded. Moreover,

$$c(n-k+1)I \preceq V_k \preceq C(n-k+1)I.$$

Using (3), compare H_k and H_{k-1} by changing separately the spatial shift, the covariance inside the Gaussian convolution, and the covariance of the outer Gaussian average. Denote the corresponding contributions by $\Delta_{\mathrm{shift},k}$, $\Delta_{\mathrm{int},k}$, and $\Delta_{\mathrm{ext},k}$.

Since $S_{k-1} - S_{k-2} = m_{k-1}$, (4) gives

$$\begin{aligned} E\|\Delta_{\mathrm{shift},k}\|_{op} &\le CE\|m_{k-1}\|^{\beta} \int_0^{\infty} e^{-2t} \|(c_t^2 V_k)^{-1/2}\|_{op}^{1+\beta} dt \\ &\le \frac{C}{1-\beta}(n-k+1)^{-(1+\beta)/2}, \end{aligned}$$

where

$$\int_0^{\infty} e^{-2t} c_t^{-(1+\beta)} dt = \frac{1}{1-\beta}.$$

For the internal covariance change, stationarity gives $V_{k-1} - V_k = \Sigma_{\infty}$. Along $A_s = c_t^2(V_k + s\Sigma_{\infty})$, the Gaussian-semigroup identity and the fourth-derivative bound yield

$$\begin{aligned} E\|\Delta_{\mathrm{int},k}\|_{op} &\le C \int_0^{\infty} e^{-2t} c_t^2 \|(c_t^2 V_k)^{-1/2}\|_{op}^3 \|\Sigma_{\infty}\|_{op}\, dt \\ &\le C(n-k+1)^{-3/2} \int_0^{\infty} e^{-2t} c_t^{-1} dt \\ &\le C(n-k+1)^{-3/2}. \end{aligned}$$

For the external covariance change, couple

$$\xi_{k-1} = \xi_k + \eta_k, \qquad \xi_k \sim N(0, V_{k+1}), \quad \eta_k \sim N(0, \Sigma_{\infty}),$$

with independent summands. Applying (4) to the additional shift $e^{-t}\eta_k$ gives

$$E\|\Delta_{\mathrm{ext},k}\|_{op} \le \frac{C}{1-\beta}(n-k+1)^{-(1+\beta)/2}.$$

Consequently,

$$E\|H_k - H_{k-1}\|_{op} \le \frac{C}{1-\beta}(n-k+1)^{-(1+\beta)/2} + C(n-k+1)^{-3/2}.$$

Finally,

$$\sum_{q=1}^{n-1} q^{-(1+\beta)/2} \le \frac{C}{1-\beta} n^{(1-\beta)/2},$$

whereas $\sum_{q\ge 1} q^{-3/2} < \infty$. Summing over k and taking the supremum over $h \in Lip_1$ proves the result. □

Proof. Retain the notation from the stationary theorem. The reward Poisson equation again gives

$$U_n = W_n + \frac{V(X_0) - V(X_n)}{\sqrt{n}}, \qquad W_n := \frac{1}{\sqrt{n}} \sum_{k=1}^{n} m_k,$$

and boundedness of V implies

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le d_{\mathcal{W}}(W_n, \Sigma_\infty^{1/2} Z) + O(n^{-1/2}).$$

Let

$$V_k := \sum_{j=k}^{n} E(\Sigma_j), \qquad D_k := V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}.$$

Geometric ergodicity gives

$$\|E(\Sigma_k) - \Sigma_\infty\|_{op} \le C\rho^{k-1}$$

for some $\rho \in (0,1)$. Therefore,

$$\frac{1}{n} V_1 = \Sigma_\infty + O(n^{-1}).$$

Since $\Sigma_\infty \succ 0$, the matrix square-root map is locally Lipschitz on the positive-definite cone. Coupling the two Gaussian vectors with the same standard normal vector therefore yields

$$d_{\mathcal{W}}\left(n^{-1/2} V_1^{1/2} Z, \Sigma_\infty^{1/2} Z\right) = O(n^{-1}).$$

Geometric convergence to the positive-definite matrix Σ_∞ also implies that, for all sufficiently large n, there are constants $0 < c < C < \infty$ such that

$$c(n-k+1)I \preceq V_k \preceq C(n-k+1)I, \qquad k = 1, \dots, n. \tag{5}$$

Indeed, $E(\Sigma_j) \succeq \frac{1}{2}\lambda_{\min}(\Sigma_\infty) I$ for all sufficiently large j, while the matrices $E(\Sigma_j)$ are uniformly bounded. For indices k preceding this fixed threshold, the positive-definite tail of the sum still contains a fixed positive fraction of the $n-k+1$ terms.

It follows from (5) that

$$\|V_k^{1/2}\|_{op} \le C(n-k+1)^{1/2}, \qquad \|V_k^{-1/2}\|_{op} \le C(n-k+1)^{-1/2},$$

and

$$\mathrm{Tr}(D_k) = \mathrm{Tr}(V_k^{-1} E(\Sigma_k)) \le \frac{C}{n-k+1}.$$

Since the martingale increments are uniformly bounded, the two Taylor remainders in Theorem 2 are therefore bounded by

$$\frac{C}{(1-\beta)^2} n^{-\beta/2}.$$

It remains to control the covariance-interaction term. Let

$$B(x) := \Sigma_\infty - v(x),$$

and let Φ be a bounded matrix-valued solution of

$$\Phi - P\Phi = B.$$

Define

$$\widetilde{\Phi}_k := \Phi(X_k) - E[\Phi(X_k)], \qquad \Gamma_k := \Phi(X_k) - E[\Phi(X_k) \mid \mathcal{F}_{k-1}].$$

The Poisson equation gives

$$B(X_{k-1}) = \Phi(X_{k-1}) - \Phi(X_k) + \Gamma_k.$$

Taking expectations and subtracting the resulting identity gives

$$B(X_{k-1}) - E[B(X_{k-1})] = \widetilde{\Phi}_{k-1} - \widetilde{\Phi}_k + \Gamma_k.$$

On the other hand, since $\Sigma_k = v(X_{k-1})$,

$$\begin{aligned} E(\Sigma_k) - \Sigma_k &= E[v(X_{k-1})] - v(X_{k-1}) \\ &= B(X_{k-1}) - E[B(X_{k-1})]. \end{aligned}$$

Therefore,

$$E(\Sigma_k) - \Sigma_k = \widetilde{\Phi}_{k-1} - \widetilde{\Phi}_k + \Gamma_k. \tag{6}$$

Thus, unlike in the stationary theorem, centering is needed because $E[\Phi(X_k)]$ may depend on k.

For fixed $h \in Lip_1$, define

$$H_k := V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2}.$$

Because H_k is $\mathcal{F}_{k-1}$-measurable,

$$E \operatorname{Tr}(H_k \Gamma_k) = 0.$$

Using (6) and summation by parts gives

$$\begin{aligned} &\sum_{k=1}^{n} E \operatorname{Tr}\left(H_k(E(\Sigma_k) - \Sigma_k)\right) \\ &= E \operatorname{Tr}(H_1 \widetilde{\Phi}_0) - E \operatorname{Tr}(H_n \widetilde{\Phi}_n) \\ &\quad + \sum_{k=2}^{n} E \operatorname{Tr}\left((H_k - H_{k-1})\widetilde{\Phi}_{k-1}\right). \end{aligned}$$

The matrices $\widetilde{\Phi}_k$ are uniformly bounded. Moreover, (5) and the bound on $A_{k,h}$ imply

$$\|H_k\|_{op} \le C(n-k+1)^{-1/2},$$

so the two boundary terms are $O(1)$.

The proof of Lemma 2 uses stationarity only through the linear covariance-sum bounds and the bounded one-step covariance increments. Here these ingredients are supplied by (5) together with

$$V_{k-1} - V_k = E(\Sigma_{k-1}), \qquad V_k - V_{k+1} = E(\Sigma_k),$$

whose right-hand sides are uniformly bounded positive-semidefinite matrices. The same spatial-shift, internal-covariance, and external-covariance estimates therefore give

$$\sup_{h \in Lip_1} \sum_{k=2}^{n} E\|H_k - H_{k-1}\|_{op} \le \frac{C}{(1-\beta)^2} n^{(1-\beta)/2}.$$

Consequently,

$$\Delta_n \leq \frac{C}{(1-\beta)^2} n^{-\beta/2}.$$

Applying Theorem 2 to W_n, comparing its Gaussian target with $\Sigma_\infty^{1/2} Z$, and then using the reward boundary term proves

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \leq \frac{C}{(1-\beta)^2} n^{-\beta/2}.$$

Choosing

$$\beta = 1 - \frac{2}{\log n}$$

for all sufficiently large n gives the stated

$$O\left(\frac{(\log n)^2}{\sqrt{n}}\right)$$

rate. Enlarging C if necessary covers the finitely many remaining values of n. □

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